Class 9 Biology Unit: 4 – Behind Movements **(Abstract)**

1. Pseudopodia, cilia, flagella, tentacles, setae, fin, tail, wings, limbs etc. are certain means of movement in various organisms.

2. Pseudopodial, cilial or flagellar movements are seen in microorganisms.
Movement of sperms, peristalsis, heart beat, blood circulation, walking, running etc are in animals.
Seed germination, growth of parts, tropic movements , nastic movements are seen in plants.

Movement of particles by osmosis, diffusion, active transport, transport of gases and nutrients, excretion of waste etc. are examples of movements common in all organisms.

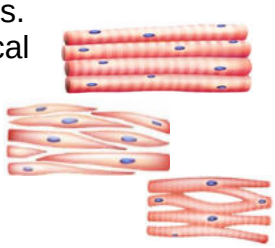
3. Cytoplasmic streaming, Pseudopodial movement (in WBC) Flagellar movement (in sperm), Ciliary movement (in the oviduct) and Muscular movement are seen in human body.

4. In humans, musclar and skeletal systems help in movement and locomotion.

5. The micro protein fibres **actine** and **myosin**, enable muscle cells to contract and relax.

6. Human body contains skeletal muscles, smooth muscles and cardiac muscles.

7. - Skeletal musles are attached to the bones and voluntary. Cells are cylindrical with striations. (When function continuously, muscle fatigue may happen).
- Smooth cells are seen in hollow organs and involuntary. Cells are spindle without striations.
- Cardiac muscles are seen only in heart wall, branched and involuntary. Cells are cylindrical with striations.



8. The antagonistic actions of both the **flexor muscle** and **extensor muscle** help us fold and strech our hands or legs.

9. Tendons connect the muscles to the bones. Ligaments connect a bone to another bone.

10.Human skeleton has a total of 206 bones. Of which, the Axial skeleton = 80 (skull 29+ sternum 1+ ribs 24+ vertebral column 26) and the Appendicular skeleton =126 (pectoral girdle 4+ forelimb 60+ pelvic girdle 2+ hindlimb 60).

11. **Periostium** is the covering membrane of a bone. **Osteoblast cells** of bones deposit minerals in the bones, make them strong and firm and also help in growth and repair.

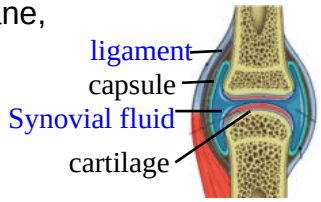
12. The connective tissue, **cartilage** is softer and more flexible than bone. Due to the absence of blood vessels, its growth is slow. Nerves are also absent in cartilage.
(cartilage also reduce friction).

13. Humans mainly have an endoskeleton of cartilage and bones. Some animals have exoskeleton with calcium carbonate shell or chitin outer covering. Invertebrates like jelly fish,hydra and snail have a hydroskeleton filled with water.

Based on evolutionary history, it can be understood that after the emergance of organisms with hydroskeleton, organisms with exoskeletons and then endoskeletons were evolved.

14. Joints are the parts where bones are connected. They help in movement and locomotion.
There are of four types of joints: pivot joint, ball & socket joint, hinge joint and gliding joint.

15. A typical joint consists of a ligament, capsule, synovial membrane, synovial fluid and cartilage.
(Both the synovial fluid and cartilage help to reduce friction between the bones).



20. Regular exercise helps to increase vital capacity of lungs and the efficiency of heart muscles, which increases blood flow and gaseous exchange.Exercise strengthens the limbs and muscles.

21. Examples of skeletal muscular disorders:
Osteoporosis (bone become porous and weak), rheumatoid arthritis (pain and swelling in the joints due to destruction of cartilages and synovial membrane), sprains (stretching or breaking of ligaments) and muscular dystrophy (muscles become weak).

22. Phototropism, geotropism, hydrotropism, haptotropism and chemotropism are **tropic movements** take place in plants.
Nastic movements, which does not follow the direction of stimuli are also seen in plants.

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Class 9 Biology Unit: 5 – Reproductive Health **(Abstract)**

Sex assigned by birth : Male, female and intersex.
Gender : Male, female, intersex and transgender.
[A transgender identify himself as non-conforming to his sex at birth and wish to live according to the gender identity in which one is convinced of]

We should ensure **gender justice** without discrimination

Adolescence is a period of physical, mental, social and sexual growth.
At this stage, puberty can be considered as the onset of ejaculation in boys and menstruation in girls.

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	Boy	Girl
Sex organ:	Testes	Ovaries
	ejaculation	*ovulation*
Gametes:	**Sperms**	**Ovum** (egg)

fertilisation (if this not occur, menstruation in girls)

Zygote (1 celled)

Morula (16-32 celled)

Implantation

Blastocyst

Gestation period (270-280 days)

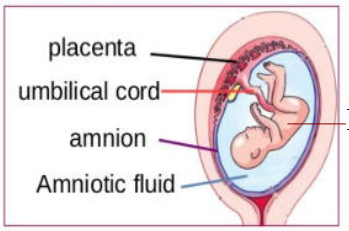
Embryo (upto 8 weeks)

Foetus

Parturition [or C-section]

Baby

Child



Foetus

Antenatal care:
- Regular check-ups, tests and vaccinations.
- Balanced diet, light exercise, rest and fun.
- Drinking adequate amount of water.
- Emotional support and Counseling
- Reduce salt,sugar, fat.
- Avoid bad habits.

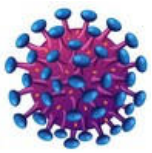
Neonatal care:
- Breastfeeding.
- Vaccination.

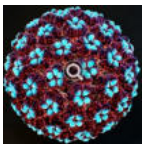
Reasons for Infertility:
Defects in sperm production, decrease in the number of sperms and their motility, ovulation disorders, blockage in the fallopian tube, hormonal imbalances such as Polycystic Ovary Syndrome (PCOS), Smoking, substance abuse, alcoholism, toxins, sexually transmitted infections, inflammation of the reproductive organs and pollution.

Treatment for infertility:
Hormone and semen test, ultrasound scanning and genetic tests can diagnose the cause of infertility. Medicines and technology like in vitro fertilisation (IVF)

Contraceptive methods:
Condoms, diaphragm, spermicides, oral contraceptive pills and intra uterine devices (IUD).
Vasectomy (cutting or tying the vas deferens) in males and tubectomy (cutting or tying the oviduct) in females

Sexually Transmitted Infections:
Virus infections: AIDS due to HIV, Genetal herpes, Human Papilloma Virus (HPV) infection,Hepatitis B.
Bacterial infections : Chlamydiasis, Gonorrhea (Gonodococci), Syphilis (Treponema).
Fungal infection: Candidiasis.
Parasite infection: Trichomoniasis.
Genital hygien is important.


HIV


HPV


Gonodococci


Treponema

Class 9 Biology Unit: 6 – CLASSIFICATION **(Abstract)**

Carl Woese introduced the level **Domains** above the level **Kingdom**. He also divided the kingdom Monera into Bacteria and Archaea (thus to form 6 kingdoms)

Domain (3)	Kingdom (6)	Phylum	Sub-Phylum	Super Class	Class
1. BACTERIA	* **Bacteria** (Prokaryotes with peptidoglycan in their cell wall)				
2. ARCHAEA	* **Archaea** (Prokaryotes which lack peptidoglycan in their cell wall)				
3. EUKARYA (eukaryotes)	* **Protista** (Unicellular eukaryotes, like amoeba, paramoecium and euglena) * **Fungi** (Unicellular/multicellular eukaryotes & are non motile heterotrophs) * **Plantae**	- **Algae** (thallus body) (No vascular tissues. eg:- Spyrogyra, Sargassum) - **Bryophyta** (Roots, stem & leaves, but no vascular tissues. eg:- Riccia, Funaria) - **Pteridophyta** (With simple vascular tissues. eg:- Lycopodium, Pteris) - **Gymnosperms** (naked seeds) (Complex vascular tissues without xylem vessels. eg:- Cycas, Pine) - **Angiosperms** (Flowering plants)			
	* **Animalia**	* **Porifera** (porous) eg:- Sponges * **Cnidaria** (with cnidoblasts) (eg:- Hydra, Jellyfish) * **Platyhelminthes** (flat worms) (eg:- Planaria, Tapeworm) * **Nematoda** (round worms) (eg:- Round worm, Hook worm) * **Annelida** (segmented) (eg:- Earthworm) * **Arthropoda** (joint footed) (eg:- Crab, Cockroach, Prawn) * **Mollusca** (soft body) (eg:- Snail, Octopus) * **Echinodermata** (spiny body) (eg:- Star fish) * **Chordata** (with notochord or vertebral column)	- **Vertebrata** (with vertebral column)	1.**Pisces** (fishes) 2. **Tetrapoda** (having 4 limbs)	- **Chondrichthyes** (cartilaginous) - **Osteichthyes** (bony fishes) 1. **Amphibia** (amphibians) 2. **Reptilia** (reptiles) 3. **Aves** (birds) 4. **Mammalia** (mammals)

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Man: Classification levels
Domain: *Eukarya*
Kingdom: *Animalia*
Phylum: *Chordata*
Sub-phylum: *Vertebrata*
Super class: *Tetrapoda*
Class: *Mammalia*
Order: *Primates*
Family: *Hominidae*
Genus: *Homo*
Species: *sapiens*

* **Viruses** are not included in any domains because of very simple structure devoid of cytoplasm, cell organelles or a nucleus and they cannot multiply without the help of a host cell.

* To classify a newly discovered organism, not only its characteristics compared to those of others, but DNA also.

* An **evolutionary tree** illustrates the evolutionary relationships between different organisms as branches on a tree. This is helpful in understanding the classification clearly.

* **DNA barcoding** is the moderntechnology that classifies organisms by comparing specific molecularsequences (codes) in DNA. This technology helps to identify species at the molecular level.